

Peer Tutoring and First-Year Programming Outcomes

Imran Hossain and Clara Whitfield

Abstract

Many first-year students fail their first programming course. We examine whether weekly peer tutoring improves examination results. Among 412 students at one university, those who attended at least eight sessions scored 7.8 points higher than matched students who attended none.

1 Introduction

Introductory programming has one of the highest failure rates of any first-year course. Universities have tried smaller classes, automated feedback and peer tutoring, in which second-year students help first-year students in small groups. Evidence for peer tutoring comes mostly from small classes in which students chose whether to take part.

This paper compares students who attended tutoring with similar students who did not, using records from three cohorts.

2 Method

We obtained anonymised records for 412 students who took the course between 2020 and 2022. Attendance at tutoring was recorded by the tutors. Each student who attended at least eight sessions was matched with a student of the same entry grade, gender and programme who attended none. The outcome is the final examination mark out of 100.

3 Results

Students who attended tutoring scored 7.8 points higher on average (95% confidence interval 4.1 to 11.5). The difference was largest for students with the lowest entry grades. The pass rate was 81% among attenders and 66% among matched non-attenders.

We did not observe any difference between cohorts taught online and those taught on campus.

4 Discussion

The size of the difference is similar to that reported for supplemental instruction in mathematics. Tutoring appears to help most the students who arrive least prepared, which suggests that it narrows rather than widens gaps between students.

These results should be read with care. The main limitation of this study is that students chose whether to attend, so motivated students may be over-represented among attenders even after matching. We did not measure prior programming experience, which could explain part of the difference. The data come from a single university, and we cannot generalize the findings to institutions with different entry requirements.

Despite this, the pattern was the same in all three cohorts, which makes a chance finding unlikely.

5 Conclusion

Weekly peer tutoring was associated with clearly better examination results in a large first-year programming course, especially for weaker students. The scheme is cheap to run because tutors are second-year students.

Further research should assign tutoring places by lottery so that the effect can be separated from motivation. We plan to follow the 2022 cohort into the second year to see whether the advantage lasts.

References

- Bennedsen, J. and Caspersen, M. (2019). Failure rates in introductory programming: 12 years later. *ACM Inroads*, 10(2), 30-36.
- Dawson, P., van der Meer, J., Skalicky, J. and Cowley, K. (2014). On the effectiveness of supplemental instruction. *Review of Educational Research*, 84(4), 609-639.